№0

$$\begin{cases} u_{tt} + u_{xx} = 0, \quad x \in \mathbb{R}^1, \ t > 0 \\ u|_{t=0} = 0 =: \varphi_0(x) \\ u_t|_{t=0} = 0 =: \psi_0(x) \end{cases}$$

$$u_0(x,t) \equiv 0$$

№k

$$\begin{cases} u_{tt} + u_{xx} = 0, \quad x \in \mathbb{R}^1, \ t > 0 \\ u|_{t=0} = 0 =: \varphi_k(x) \\ u_t|_{t=0} = \frac{1}{k}\cos kx =: \psi_k(x) \end{cases}$$

$$|\psi_k(x) - \psi_0(x)| = \frac{1}{k}|\cos kx| \le \frac{1}{k}$$

$$\psi_k(x) \underset{k\to\infty}{\overset{\mathbb{R}^1}{\rightrightarrows}} \psi_0(x)$$

$$u_k(x,t) = \frac{1}{k^2}\operatorname{sh} kt \cos kx = \frac{1}{2k^2}\left(e^{kt} - e^{-kt}\right)\cos kx$$

$(\hat{x}, \hat{t})$

$$|u_k(\hat{x},\hat{t}) - u_0(\hat{x},\hat{t})| = \left|\frac{1}{2k^2}\left(e^{k\hat{t}} - e^{-k\hat{t}}\right)\cos k\hat{x}\right| \ge \varepsilon_0$$

$$\hat{x} = 0, \quad \hat{t} = \frac{1}{\sqrt{k}}$$

$$\left|\frac{1}{2k^2}\left(e^{\sqrt{k}} - e^{-\sqrt{k}}\right)\right| = \frac{1}{2}\frac{1}{k^2}e^{\sqrt{k}}\underbrace{\left(1 - \frac{1}{e^{2\sqrt{k}}}\right)}_{\ge \frac{1}{2}} \ge \frac{1}{4}\left(\frac{1}{k^2}e^{\sqrt{k}}\right) \ge \varepsilon_0$$

нач. с нек. k

$x = (x_1, x_2, x_3)$

$u(x,t)$

$$\begin{cases} u_{tt} - a^2\Delta u = 0, \quad x \in \mathbb{R}^3, \ t > 0 \\ u|_{t=0} = \varphi(x), \quad x \in \mathbb{R}^3 \\ u_t|_{t=0} = \psi(x), \quad x \in \mathbb{R}^3 \end{cases}$$

$$\begin{cases} x = x \\ \tau = at \end{cases}$$

$\tilde{u}(x,\tau)$

u_{x_j}	$u_{x_jx_i}$	u_t	u_{tt}
$\parallel$	$\parallel$	$\parallel$	$\parallel$
$\tilde{u}_{x_j}$	$\tilde{u}_{x_jx_i}$	$\tilde{u}_\tau \cdot a$	$a^2\tilde{u}_{\tau\tau}$

$$a^2 u_{\tau\tau} - a^2\Delta u = 0, \quad x \in \mathbb{R}^3, \ \tau > 0$$

$$\begin{cases} u_{\tau\tau} - 1\Delta u = 0 \\ u|_{\tau=0} = \varphi(x) \\ u_\tau|_{\tau=0} = \frac{1}{a}\psi(x) \end{cases}$$

$\varphi(x) \in C^3(\mathbb{R}^3)$,

$\psi(x) \in C^2(\mathbb{R}^3)$

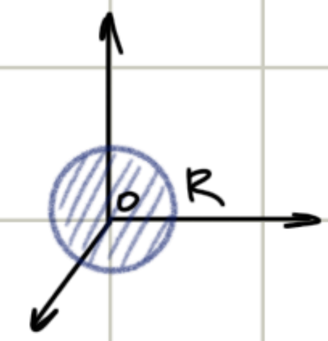
$$u(x,t)=\frac{1}{4\pi a^2 t}\iint\limits_{|y-x|=at}\psi(y)\,dS_y+\frac{\partial}{\partial t}\left[\frac{1}{4\pi a^2 t}\iint\limits_{|y-x|=at}\varphi(y)\,dS_y\right]$$

$\varphi(x)\equiv 0$ в $\mathbb{R}^3$

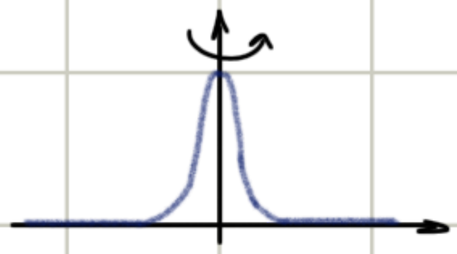
$\psi(x)\to C^\infty(\mathbb{R}^3)$

$\psi(x)\to \exists R:\ \psi(x)=0\quad \forall x:|x|\geq R$

$\psi(x)\to \psi(x)\geq 0\quad \forall x\in\mathbb{R}^3$



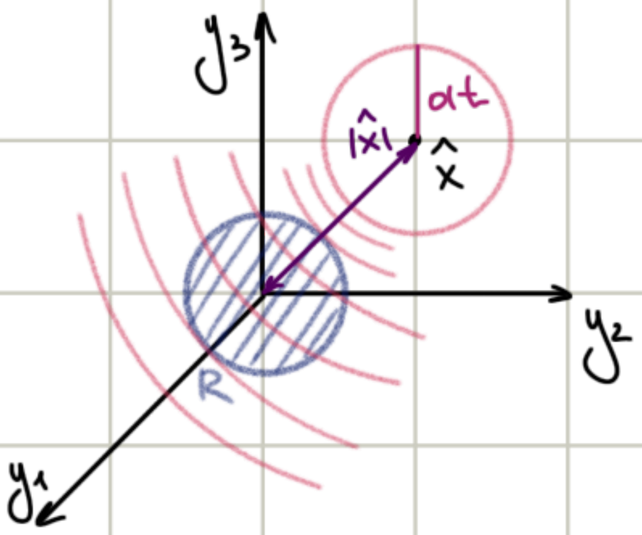
$\mathbb{R}^2$



$\mathbb{R}^1$

$$u(x,t)=\frac{1}{4\pi a^2 t}\iint\limits_{|y-x|=at}\psi(y)\,dS_y=\begin{cases}0, & \text{если } at\leq|\hat{x}|-R\\ \text{"не ноль"}, & \text{если } |\hat{x}|-R<at<|\hat{x}|+R\\ 0, & \text{если } at\geq|\hat{x}|+R\end{cases}$$

$\hat{x}$ t



$at-R\leq|x|\leq at+R$

$\{at-R=|x|\}$ $\{|x|=at+R\}$

физ. смысл a - скорость распр. звуковой волны

$x=(x_1,x_2)$

$u(x,t)$

$\varphi(x)\in C^3(\mathbb{R}^2)$, $\psi(x)\in C^2(\mathbb{R}^2)$

$$\begin{cases}u_{tt}-a^2\Delta u=0, & x\in\mathbb{R}^2,\ t>0\\ u|_{t=0}=\varphi(x), & x\in\mathbb{R}^2\\ u_t|_{t=0}=\psi(x), & x\in\mathbb{R}^2\end{cases}$$

$$u(x,t)=\frac{1}{2\pi a}\iint\limits_{|y-x|<at}\frac{\psi(y)\,dy}{\sqrt{(at)^2-|y-x|^2}}+\frac{\partial}{\partial t}\left[\frac{1}{2\pi a}\iint\limits_{|y-x|<at}\frac{\varphi(y)\,dy}{\sqrt{(at)^2-|y-x|^2}}\right]$$

$y=(y_1,y_2)\to (y_1-x_1)^2+(y_2-x_2)^2<(at)^2$

$\to |y-x|<at$

зададим точечное возмущение

$\varphi(x)\equiv 0$ в $\mathbb{R}^2$

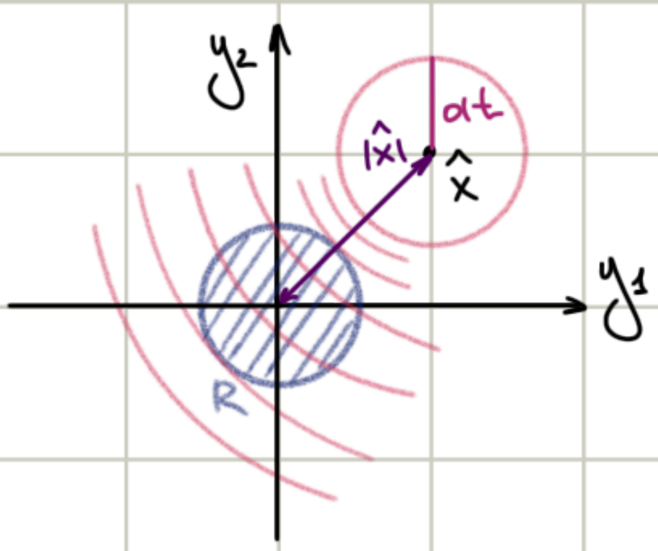
$\psi(x)\to C^\infty(\mathbb{R}^2)$

$\psi(x)\to \exists R>0:\ \psi(x)=0\quad \forall x:|x|\geq R$

$\psi(x)>0\quad \forall x:|x|<R$

$$u(\hat{x},t)=\frac{1}{2\pi a}\iint\limits_{|y-\hat{x}|<at}\frac{\psi(y)\,dy}{\sqrt{(at)^2-|y-\hat{x}|^2}}=\begin{cases}0, & at\leq|\hat{x}|-R\\ \text{"не ноль"}, & at>|\hat{x}|-R\end{cases}$$

$\hat{x}$?t



теперь имп. по кругу

$u(\hat{x},t)>0 \quad \forall\, t: at>|\hat{x}|-R$

$\lim\limits_{t\to+\infty} u(\hat{x},t)=0$

$x=(x_1,x_2,x_3)$

$u(x,t)$

(1) $u_{tt}-\Delta u = F(x,t),\ x\in\mathbb{R}^3,\ t>0$

(2) $u|_{t=0}=\varphi(x),\ x\in\mathbb{R}^3$

(3) $u_t|_{t=0}=\psi(x),\ x\in\mathbb{R}^3$

$\exists\ \varphi(x)\in C^3(\mathbb{R}^3)$

$\psi(x)\in C^2(\mathbb{R}^3)$

$F(x,t)\in C^2(x\in\mathbb{R}^3,\ t\geq 0)$

рем $\exists!$ и удд. ф-й Кирхгофа

(1$_0$) $u_{tt}-\Delta u = 0,\ x\in\mathbb{R}^3,\ t>0$

(2) $u|_{t=0}=\varphi(x),\ x\in\mathbb{R}^3$

(3) $u_t|_{t=0}=\psi(x),\ x\in\mathbb{R}^3$

(1) $u_{tt}-\Delta u = F(x,t),\ x\in\mathbb{R}^3,\ t>0$

(2$_0$) $u|_{t=0}=0,\ x\in\mathbb{R}^3$

(3$_0$) $u_t|_{t=0}=0,\ x\in\mathbb{R}^3$

1 шаг

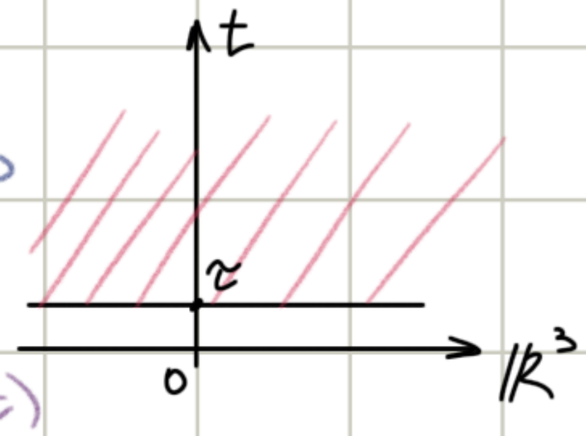
$\tau:\tau\geq 0$

$v(x,t)\longmapsto v(x,t,\tau)$

$$\begin{cases} v_{tt}-\Delta v=0,\ x\in\mathbb{R}^3,\ t>\tau \\ v|_{t=\tau}=0,\ x\in\mathbb{R}^3 \\ v_t|_{t=\tau}=\underbrace{F(x,\tau)}_{C^2},\ x\in\mathbb{R}^3 \end{cases}$$

$v(x,t,t)=0$

$v(x,\tau,\tau)=0$



$v_t'(x,t,t)=F(x,t)$

$v_t'(x,\tau,\tau)=F(x,\tau)$

$\xi=t-\tau$

$v_{\xi\xi}-\Delta v=0,\ x\in\mathbb{R}^3,\ \xi>0$

$v|_{\xi=0}=0,\ x\in\mathbb{R}^3$

$v_\xi|_{\xi=0}=\underbrace{F(x,\tau)}_{C^2(\mathbb{R}^3)},\ x\in\mathbb{R}^3$

2 шаг

решение (1), (2$_0$), (3$_0$)

$u(x,t)=\int_0^t v(x,t,\tau)\,d\tau$

(2$_0$) очевидно выполняется

(30) $u_t(x,t) = v(x,t,t) + \int_0^t v'_t(x,t,\tau)\,d\tau$ ($v(x,t,t) \to 0$)

$u_t|_{t=0} = 0$, т.е. выполнено

$u_{tt}(x,t) = (u_t(x,t))'_t = v'_t(x,t,t) + \int_0^t v_{tt}(x,t,\tau)\,d\tau$ $\Rightarrow$ $u_{tt} = F(x,t) + \Delta u$

$F(x,t)$ — $\Delta v(x,t,\tau)$, т.к. $v_{tt} - \Delta v = 0$

(1) выполнено

$u_{x_1x_1}(x,t) = \int_0^t v_{x_1x_1}(x,t,\tau)\,d\tau$

$\Delta(u(x,t)) = \int_0^t \Delta(v(x,t,\tau))\,d\tau$

принцип Дюамеля